

Does Peer Review Need to Change? A Panel on Reporting Standards and Checklists in the Age of AI

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Abstract

Many scientific fields of study use formally established reporting standards to foster research and experimental design, transparency, replicability, peer review, and student training. Examples include CONSORT in medicine, the What Works Clearinghouse in education, JARS in psychology. Such standards yield agreement on study reporting and evaluation, even if using different methodologies. CHI has not adopted reporting standards. Like other fields, CHI has seen an increased number of low-quality submissions and reviews fueled by AI. This panel's objective is to discuss advantages and barriers of adopting reporting standards for SIGCHI. Panelists include representatives with significant experience creating, adopting and operationalizing reporting standards in adjacent fields: software engineering, CS education, and Programming Languages. The panel will include an overview of the history of reporting standards, a live demo of a standards-based peer review system, discussions of opportunities, challenges, limitations for SIGCHI reporting standards, and an interactive discussion between attendees and panelists.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

evidence standards, reporting templates, review checklists

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1 Moderator, Organizers and Panelists

Andrew Duchowski (moderator) cofounder of ACM ETRA, the Symposium on Eye Tracking Research & Applications, has co-authored numerous publications on eye-tracking work most of which involved human participation, empirical methodology, and statistical analyses. As a regular attendee of CHI, he has a deep understanding of the concerns and norms of the CHI community. **Andreas Stefik** (organizer) has spearheaded several efforts at bringing reporting standards to multiple areas of Computer Science, including co-organizing two Dagstuhl seminars¹ on this topic. He will bring to the panel a wealth of historical knowledge about reporting standards, including their medical origins. **Paul Ralph** (organizer) is founding editor-in-chief of the ACM SIGSOFT Empirical Standards for Software Engineering (SE) Research, the most comprehensive evidence and reporting standards available for any computing subdiscipline. He will present his interactive standards-based reviewing tools, which are used by thousands of researchers, and speak to how easily they can be adapted for human-computer interaction (HCI) research. **Alan Dix** (panelist) co-authored a CHI '20 paper arguing for pre-registration to combat HARKing [11] while **Krzysztof Krejtz** (panelist) has substantial expertise in empirical methodology and statistical analyses regarding the quality (or lack



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¹<https://www.dagstuhl.de/18061>, <https://www.dagstuhl.de/22442>

thereof) in CHI publications and can explain the many benefits of reporting standards for statistical education and reporting. Panelists Dix and Krejtz will comment on the intersection of standards with HARKing, pre-registration and statistical rigor. As Editor-in-Chief of *Computers & Graphics*, **Joaquim A. Jorge** (panelist), UNESCO Chair on AI & XR, will give a journal editor's perspective on how poor reporting affects editorial quality control, reviewer workload, and AI-generated "slop" reviews. Meanwhile, **Brad Myers** (panelist) has published diverse qualitative and quantitative studies in HCI, SE and programming languages (PL), including over 100 at CHI and 29 at ICSE. He will comment on potential pitfalls in applying reporting standards to evaluative research in HCI, software engineering, and programming languages. In contrast, **Rina Wehbe** (panelist), who has published influential papers at ACM conferences including CHI and CHIPLAY, will speak to the impact of reporting standards on early-career researchers and mixed-methods research.

2 Introduction and Background

2.1 Quality in Science

The integrity of research and publication processes must up be upheld to guarantee the trustworthiness, reproducibility, and freedom from manipulation or bias of empirical findings. This includes ethics, accuracy, consistency, and reliability of data and their analysis [29], the rigor of study design, and the quality of research reporting.

Across diverse fields, scientists are increasingly concerned about the quality of gathered evidence [14, 20]. This concern stems from many reasons including: a lack of procedures to detect fraud; challenges in replication [4, 32]; lack of pre-registration [21]; p-hacking (AKA data dredging) [7]; HARKing (Hypothesizing After the Results are Known) [11]; and poor transparency rooted in failing to adhere to exacting reporting standards [35].

Computer Science in general, and HCI in particular, are not immune to any of these challenges and faces its own unique difficulties in addressing them. Although the ACM Publications Board recently updated the ACM Authorship Policy and the ACM Peer Review Policy to address the use of AI in the authorship and peer review processes,² informal complaints and inconsistencies remain. We are not aware of any formal studies of CHI review procedures and no specific standard has been put forth nor adopted by the CHI community. In contrast, multiple years-long investigations into SE [17, 33] and programming languages [15, 30, 31] have furnished compelling evidence that authors in these fields fail to present rigorous evidence in their publications and lack basic checks and balances like "gathering data" [33], "having a control group" [15, 25], or testing people "other than the authors of the publication" [17]. Researchers have raised concerns about the quality of research in Computer Science Education [10] and specifically the lack of replication [1]. Meanwhile, issues of evidence quality have also motivated political change and are beginning to limit the ability of computer scientists to win government grants. For example, the Every Student Succeeds Act in the United States places empirical studies into tiers of evidence, which automatically discount many papers in our field because of our lack of reporting standards [36].

²<https://www.acm.org/publications/policies/>

2.2 Scientific Quality Control in the Age of AI

The primary quality control mechanism used in the sciences is scholarly peer review. Scholarly peer review "is an organized method for evaluating scientific work ... to certify the correctness of procedures, establish the plausibility of results, and allocate scarce resources such as journal space, research funds, recognition, and special honor" [9]. Peer review is extremely important because it determines what research is funded, what is published where, who is hired, promoted, and tenured [6, 19]; as well as how universities and scientists are ranked [5, 22]. Ziman [38] therefore called scholarly peer review the "lynchpin" about which all science (social and otherwise) pivots. Peer review also signals the legitimacy or validity of our work to policymakers, educators, students, journalists and the wider public [9]. And, of course, all of the CHI conference (even this panel!) relies on peer review. Many venues, including CHI, are seeing an explosion of submissions—including some generated by AI [8]—which increases strain on the peer review process.

While most researchers believe peer review is effective [37], decades of research consistently shows that "peer review is prejudiced, capricious, inefficient, ineffective, and generally unscientific" [26]. Study after study, from lab experiments to longitudinal data analyses to observational field studies consistently find that peer review is prejudiced (against women, racial minorities, new ideas, results that contradict reviewers' expectations, qualitative research, etc.), focused on style over substance, irrational, unreliable, opaque, inefficient, and mostly random. Bad peer review causes emotional injury, burnout, and discrimination against protected groups, makes science inefficient, creates publication bias, and allows fundamentally flawed research to influence policy.

More recently, editors and program chairs have seen an explosion of AI-slop (low-quality, AI-generated or AI-assisted) peer reviews. Reviewers are uploading manuscripts to AI chatbots and either:

- (1) asking the chatbot to write the review without the reviewer ever reading the paper;
- (2) writing a superficial review based on a chatbot's superficial summary, having never read the manuscript; or
- (3) skimming the paper and having the chatbot transform brief notes into a substantive review, which typically does not engage with the methodological substance of the paper.

The depth of this crisis for research quality control cannot be overstated. Large Language Models (LLMs) cannot reliably differentiate methodologically sound research from well-written tosh.

2.3 Reporting Standards and Evidence Standards

An *evidence standard* is model of a scientific community's expectations for a specific kind of research (e.g., HCI researcher's expectations for meta-analytic systematic reviews). Evidence standards may focus on the information an article should report, how research should be done, or the nature of evidence needed to take certain actions. When the focus is on what to report, we call it a *Reporting Standard*. The Equator Network³ aggregates hundreds of evidence and reporting standards including the CONSORT statement for reporting randomized controlled trials [12], the PRISMA statement

³equator-network.org

for reporting systematic reviews [23], and the American Psychological Association's Journal Article Reporting Standards [2, 18]—APA JARS. Both the CONSORT and PRISMA statements appear to have improved methodological soundness and reporting [16, 24, 34].

Although creating and complying with some of these standards may seem daunting or unnecessary for Computer Science and HCI, many papers discuss them for programming languages [15, 30, 31], computational methods [32], and SE [13, 14, 33]. Transactions on Computing Education (TOCE) went further and formed a reporting standards committee led by one of our panelists, analyzed reporting standards across other fields, and put the change to a vote of the Editorial board. Ultimately, the APA JARS standard was chosen due to its flexibility in empirical methodologies and reporting. For papers using the standard, authors and reviewers benefit. One crucial idea is that since authors and reviewers know ahead of time what to report and how to report it, there is a shared mental model of expectations.

Meanwhile, a team of SE researchers led by one of our panelists generated evidence standards for 18 common methodologies, most of which are also used in HCI research (e.g., controlled experiment, questionnaire survey, case study) and an interactive system for supporting peer review using these checklists [27]. The ACM SIGSOFT Empirical Standards for SE research [28]⁴ have been used by over 10,000 scholars from 90 countries and are recommended by many prestigious journals and conferences. Expectations are divided into “essential”, “desirable”, or “optional”, and editors (or program chairs) can decide which, whether or how many of the desirable and optional expectations must be met. The system transforms lists of attributes from relevant standards into a single, interactive form, which could be used as either a “pre-submission form” to help authors verify that their manuscripts meet all expectations, or a single- or multi-phase “review form” to guide reviewers [3].

Such standards are used in multiple ways: to review a paper (especially one with an unfamiliar method); to highlight the quality of one's own research (e.g., our study meets all the criteria listed in

⁴<https://www2.sigsoft.org/EmpiricalStandards/>

yes no

- ☒ states a purpose, problem, objective, or research question
- ☒ explains why the purpose, problem, etc. is important (motivation)
- ☒ defines jargon, acronyms and key concepts
- ☒ names the methodology or methodologies used
- ☒ methodology is appropriate (not necessarily optimal) for stated purpose, problem, etc.
- ☒ describes in detail what, where, when and how data were collected (see the [Sampling Supplement](#))
- ☒ describes in detail how the data were analyzed
- ☐ states formal hypotheses

is the deviation reasonable? ☐ yes ☒ no

Please indicate the type of unreasonable deviations. (Pick the largest number that applies.)

☒ type 1 ☐ type 2 ☐ type 3 ☐ type 4

How can this problem be addressed

For the ANOVA, something like "H1: Visible scroll bars increase user satisfaction"

- ☒ uses two-sided hypotheses OR justifies use of one-sided hypotheses based on face validity or previous work
- ☒ describes the dependent variable(s) and justifies how they are measured (including units, instruments)
- ☒ describes the independent variable(s) and how they are manipulated or measured

Figure 1: Excerpt from ACM SIGSOFT Empirical Standards Interactive Review Checklist for Experiments with Human Participants

- ☐ names the methodology or methodologies used
- ☐ methodology is appropriate (not necessarily optimal) for stated purpose, problem, etc.
- ☐ describes in detail what, where, when and how data were collected (see the [Sampling Supplement](#))
- ☐ describes in detail how the data were analyzed
- ☐ justifies the selection of the case(s) or site(s) that was(were) studied
- ☐ describes the site(s) in rich detail
- ☐ the site is a real (physical or virtual) place, not a toy example or imaginary world
- ☐ reports the type of case study
- ☐ describes data sources (e.g. participants' demographics and work roles)
- ☐ defines unit(s) of analysis or observation
- ☒ this is an attention check item. Select 'no'.

Figure 2: Excerpt from ACM SIGSOFT Empirical Standards Interactive Review Checklist for Case Studies

standard S); as a pre-submission checklist; and as a training aid for graduate students and research methods courses [27].

3 Main Topics of Discussion

The main discussion topics are reporting and evidence standards: what they are; how they are created; their operationalization as tools for scholarly peer review; and how these tools improve review quality, reduce AI-slop reviews, and could be adopted across SIGCHI and related journals and conferences. Figs. 1 and 2 show examples from such a tool. Note how different research methods have significantly different questions to protect methodological and epistemological pluralism. These topics are fundamentally important because they impact how scholars will report their work and what data they will need to gather to make their work acceptable for publication at these venues.

3.1 Controversial Positions

The panelists will advance several controversial and provocative positions including the following. Not all panelists agree with all these positions, so lively debate is expected.

- (1) CHI needs comprehensive reporting standards to improve paper and peer review quality. Standards have been effective in similar fields and can work here too.
- (2) CHI should require reporting standards (vs. just recommending them).
- (3) Reporting standards can benefit the public as evidence from similar methodologies is reported more consistently.
- (4) Scientific fields need to track data about the quality, or lack thereof, of the evidence in the field.
- (5) CHI should adopt pre-registration (like psychology/medicine).
- (6) Current peer reviewing is essentially essay-writing. Pilots and mechanics do not assess the flightworthiness of airplanes by writing essays about them; they complete pre-flight checklists. Practically all quality control, from surgeons and building inspectors, restaurant inspectors, and lab technicians, relies on checklists, not essays.
- (7) Interactive checklists (as in the SE standards) reduce reviewing time and effort.
- (8) HCI researchers know their field best and are better suited to arguing, debating, and instituting high quality standards than politicians that might prefer to impose their own.
- (9) Strict standards will marginalize design/critical studies.

Table 1: Panel agenda and timeline.

Time	Topic
5 min	Moderator Duchowski will welcome the guests, introduce the panelists, briefly describe reporting standards and why CHI needs them.
10 min	Co-organizer Stefik will recount the history of reporting standards, their current status of adoption in Computer Science; and the need to adopt formal standards before they are imposed by government.
10 min	Co-organizer Ralph will provide a live demo of the SIGSOFT interactive standards-based peer review system, showing how it resists AI-slop reviewing and describing how it could be adapted for HCI research.
50 min	Interactive conversation-in-motion between attendees and panelists, moderated by Duchowski. While the conversaton's direction will depend on the attendees, we expect some discussion of adoption pathways for CHI venues, including incentives, tooling, and cross-community alignment.

- (10) We do not have to start from scratch. The SE evidence standards and open source reviewing tools can be adapted and extended for HCI research, especially since some HCI methods are already included.
- (11) Using an LLM to generate a superficial peer review essay is easy. Using an LLM to interact with an interactive, standards-based review checklist is difficult, and such systems can include AI detection mechanisms (e.g., invisible questions, prompt injection, time checks).
- (12) Reporting standards and evidence standards have many more benefits, especially when operationalized as interactive pre-submission and peer review checklists.

3.2 Benefits

A peer review system operationalizing reporting and evidence standards as interactive checklists simultaneously addresses a host of problems. The third author's team conducted a randomized controlled experiment at an A-ranked SE conference, in which reviewers were randomly divided into two groups: one using a typical conference review process; the other using evidence standards checklists. The checklist group had significantly improved inter-reviewer reliability and engaged more with methodology (over style), without harming attitudes toward the review process or increasing workload. In the longer term, agreeing on standards in advance helps authors and reviewers construct shared norms for what is considered reasonable for publication, and the checklist system simultaneously helps authors comply with—and discourages reviewers from inventing criteria that depart from—community norms and published methodological guidance. The standards help train graduate students and help reviewers more competently review papers using less familiar methods and statistics. Meanwhile, the system discourages AI-slop reviewing and enables limited, evidence-based AI supported pre-review checks.

4 Panel Format & Agenda

The panel agenda is structured as shown in Table 1.

5 Plan for Engaging Audience Members

Once the organizers have set the stage with a short history of reporting standards and demo of standards-based reviewing tools, we will engage the audience using an interactive conversation-in-motion format that blends live polling, real-time audience prompts, and rotating mini-dialogues among panelists. We will use a web app

such as mentimeter(.com) where attendees can submit and upvote questions, then live-react to the ensuing discussion. The moderator will invite the audience to respond to a live poll that shapes the panel's direction, allowing attendees to choose collectively which themes to pursue. Throughout, roving microphones will be available for attendees who wish to join the discussion. This will create a more emergent, interactive exchange of ideas.

6 Why this Panel will be Exciting, Enjoyable, Well-Attended, and Relevant

Many scholars across many subdisciplines of Computer Science and related fields are upset about peer review. Practically every research-active scholar experiences not only review comments and scores that seem unjust, destructive, philosophically inappropriate, or factually wrong but also intransigent reviewers unwilling to update their perspectives no matter the facts, data, arguments or tone of the author's response. Meanwhile, practically every reviewer encounters entirely avoidable errors in research design, missing crucial information, or incorrect analyses. Editors and program chairs, in turn, often feel unable to intervene effectively without appearing biased or damaging relationships. Thus, as soon as one starts talking about changing a conference's peer review process, researchers will show up. The third author's presentation of the results of the review checklist experiment at the *International Conference on Evaluation and Assessment in Software Engineering* was the best attended session. Everyone wanted to know how the checklists affected peer review. The panel will be exciting and enjoyable because modernizing a century of peer review practice is highly controversial, and our conversation-in-motion format will help keep the audience engaged.

7 Summary

To summarize, scientific communities can collectively construct reporting standards to express their expectations for publication and review. These standards *create* consensus, improve paper and peer review quality, protect methodological pluralism, and facilitate graduate education. Operationalizing the standards as interactive checklists can streamline peer review while inhibiting inappropriate use of AI. The open source SIGSOFT review checklist system can be adapted for HCI research. The objectives of this panel are therefore to raise awareness around reporting standards; extend the community of scholars from SE, human factors, and CS Education to HCI, and to facilitate collaboration between these fields in

adopting/adapting reporting standards and modernizing peer review. This requires sensitivity to the expertise needed to formulate standards that make sense to the researchers who will ultimately use them.

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